

SalaryLens

An Intelligent Resume-Based Salary Prediction System Using Stacking Ensemble Learning

Stacking Ensemble

Resume Parsing

XGBoost

Feature Engineering

LLM Integration

Career Analytics

97.02%

R² Score

91.29%

±10% Accuracy

01

Objectives & Problem Statement

- Address information asymmetry in salary negotiation — job seekers lack data-driven market value insights
- Achieve high prediction accuracy across diverse job roles and experience levels using ML
- Capture non-linear relationships between resume attributes and compensation through 104 engineered features
- Deliver real-time predictions via an accessible web interface with interactive what-if analysis
- Integrate LLM-based automated resume parsing to extract structured data from PDF/DOCX/TXT files

02

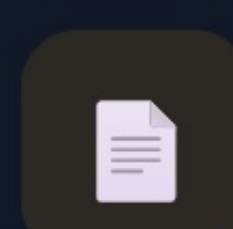
Abstract

SalaryLens is an intelligent web-based system that predicts annual salaries from resume data using advanced machine learning. The system employs a **stacking ensemble architecture** combining XGBoost, Random Forest, Extra Trees, and Neural Networks with Ridge regression as a meta-learner.

Trained on **29,850 resume records** with 104 engineered features, the model achieves 97.02% R² score and 91.29% prediction accuracy within ±10% margin. LLM-based parsing via Groq + Llama 3.1 achieves 85% extraction accuracy in under 3 seconds.

The deployed web application provides **real-time predictions, market comparisons, and scenario analysis** — a practical tool for job seekers, recruiters, and HR professionals.

03 - PROPOSED METHODOLOGY

**Resume Upload**
PDF / DOCX / TXT input**LLM Parsing**
Groq + Llama 3.1 70B
85% accuracy**Feature Engineering**
15 raw → 104 features**Stacking Ensemble**
4 base models +
Ridge meta-learner**Salary Prediction**
Real-time ±12% CI**XGBoost**1000 estimators · lr: 0.02 · R²: 96.98%**Random Forest**400 estimators · Baseline · R²: 92.64%**Extra Trees**400 estimators · Extreme randomness · R²: 94.66%**Neural Network**256–128–64 layers · R²: 94.63%

Ridge Regression Meta-Learner

3-fold cross-validation · $\alpha = 5.0$ · Learns optimal combination → Final R²: **97.02%**

04 - KEY RESULTS

97.02%

R² Score

91.29%

±10% Accuracy

\$7,071

Mean Abs. Error

29,850

Training Records

MODEL COMPARISON

SalaryLens ★	97.02%
XGBoost	96.98%
Extra Trees	94.66%
Neural Net	94.63%
Random Forest	92.64%

05 - TOP FEATURE IMPORTANCES

Seniority Rank	12.3%
Years of Experience	10.8%
Job Title Market Value	9.5%
Seniority × Experience	8.7%
Education Rank	7.2%
Industry Market Value	6.9%
Skills × Experience	6.1%
Achievement Score	5.4%
High-Paying Location	4.8%
Company Size	4.3%

06 - NOVELTY & CONTRIBUTION

- Stacking ensemble achieving **97.02% R²**, surpassing all prior individual-model salary prediction approaches
- Comprehensive feature engineering pipeline generating **104 features** — 7.79% R² boost over raw features
- LLM-based resume parsing via **Groq API + Llama 3.1 70B** achieving 85% extraction in under 3 seconds
- Interactive **What-If analysis** with 5 career scenarios: skill upgrades, relocation, promotion, education, growth
- Deployed production-ready **web application** with real-time prediction, market comparison, and job search integration

07 - APPLICATIONS

- Job Seekers**
Data-driven salary benchmarking before negotiation
- Recruiters & HR**
Benchmark competitive compensation packages
- Career Planners**
Simulate career decisions with what-if scenarios
- Research**
Empirical salary modeling for labor economics

08 - CONCLUSION & FUTURE WORK

SalaryLens demonstrates that stacking ensemble learning with comprehensive feature engineering significantly outperforms individual models, achieving state-of-the-art performance with practical usability through an interactive web application.

- Real-world validation on live job platform data
- LIME/SHAP explainable AI integration
- Geographic expansion with currency conversion
- Fairness analysis and bias mitigation
- Mobile application development

09 - IMPACT

- Real-Time Speed**
Sub-1-second salary prediction from resume upload
- Superior Accuracy**
8% improvement over prior state-of-the-art systems
- Transparent Insights**
Feature importance reveals true salary drivers
- Accessible Design**
WCAG AA compliant, keyboard and screen reader ready